

How transparency affects algorithmic advice utilization: The mediating roles of trusting beliefs

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ABSTRACT

Although algorithms are increasingly used to support professional tasks and routine decision-making, their opaque nature invites resistance and results in suboptimal use of their advice. Scholars argue for transparency to enhance the acceptability of algorithmic advice. However, current research is limited in understanding how improved transparency enhances the use of algorithmic advice, such as the differences among various aspects of transparency and the underlying mechanism. In this paper, we investigate whether and how different aspects of algorithmic transparency (performance, process, and purpose) enhance the use of algorithmic advice. Drawing on the knowledge-based trust perspective, we examine the mediating roles of trusting beliefs in the relationships between transparency and the use of algorithmic advice. Using the “judge-advisor system” paradigm, we conduct a $2 \times 2 \times 2$ experiment to manipulate the three aspects of transparency and examine their effects on the use of algorithmic advice. We find that performance and process transparency promote the use of algorithmic advice. However, the effect of process transparency gets attenuated when purpose transparency is high. Purpose transparency is only useful when process transparency is low. We also find that while all three aspects of transparency facilitate different trusting beliefs, only competence belief significantly promotes the use of algorithmic advice. It also fully mediates the facilitating effects of performance and process transparency. This study contributes to the emerging research on algorithmic decision support by empirically investigating the effects of transparency on the use of algorithmic advice and identifying the underlying mechanism. The findings also provide practical guidance on how to promote the acceptance of algorithmic advice that is valuable to both individual users and practitioners.

1. Introduction

With advances in artificial intelligence, algorithms – a set of procedures that computers can follow to perform a task [1] – are increasingly called upon to make informed decisions [2–5]. They have demonstrated decision-support capabilities in several fields [6–8], such as medical diagnosis [9,10] and criminal sentencing [6], as well as routine decision-making, such as product recommendations [11–13]. However, their potential to improve decision quality depends substantially on people’s willingness to use them. While people seek advice from others when making decisions, they can exhibit resistance to advice from algorithms [1,14,15]. Compared to human advice, algorithmic advice is often more seriously discounted and less likely to be relied on [1,9,14]. As observed in various contexts, such as employee selection [16] and evaluation [17], fraud prediction [18], service choice [19], and

moral decisions [6,20], people do not rely on advice from algorithms [21]. The underuse of algorithmic advice prevents its true potential to improve decision-making from being realized [22].

Why are people reluctant to rely on algorithmic advice? Scholars contend that algorithms, being black boxes, are not trustworthy [23–25]. While their advice may be superior, it is difficult to pinpoint how they arrive at a particular decision [26]. The underlying process that governs algorithmic decision-making is unknown [26], and people therefore exhibit aversion to such black-box advice [27,28]. Accordingly, several studies suggest that increasing transparency (i.e., the presence of information related to algorithms) should improve people’s attitudes towards algorithmic advice [27,28]. This idea is also supported by practical evidence. A field study suggests that workers are more likely to comply with algorithms if they know some information about algorithms’ working processes [29]. Another study reports that decision-

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makers also give positive feedback if the principles of algorithms they use to make decisions are transparent. This study surveyed 450 undergraduate students' attitudes towards algorithmic advice in managerial decisions and found that about 94% of the students wanted to know the algorithm's procedure before relying on it to make a decision [26]. Thus, it appears that increasing transparency will help to promote the use of algorithmic advice.

Although some empirical studies have paid attention to the effect of transparency [11,26,30], there are two main limitations. First, most studies on algorithmic transparency have focused on a singular aspect of transparency, only highlighting the impact of either the performance [1,9,30] or process transparency [11,26,31]. These studies are scattered and have not holistically examined the impact of various aspects of algorithmic transparency. Purpose transparency, which involves the presence of purpose information in an algorithm, is also an important aspect of transparency [32] and should be empirically examined. Second, there is a lack of clear understanding regarding why increasing transparency promotes the use of algorithmic advice. The mechanism underlying this promoting effect remains unclear, which constitutes a significant gap in the current literature.

Considering these limitations, this paper aims to holistically examine the effect of transparency on people's use of algorithmic advice and explore the underlying mechanism. Specifically, we put forward two research questions: (a) how do different aspects of transparency (i.e., performance, process, and purpose) affect the use of algorithmic advice? and (b) what is the underlying mechanism that drives the promoting effect of transparency? To answer these questions, we conducted an experiment with 443 participants, using the "judge-advisor system" paradigm [1,33,34]. Following previous research [26,30], we manipulated the three aspects of transparency through explanations and investigated their effects on the use of algorithmic advice. Our findings suggest that performance and process transparency generally promote the use of algorithmic advice. Purpose transparency is only useful when process transparency is low, and the effect of process transparency gets weakened when purpose transparency is high. To explain the underlying mechanism, we drew on the knowledge-based trust perspective [35–37] and treated three trusting beliefs (i.e., competence, integrity, and benevolence) as the mediators through which algorithmic transparency affects the use of algorithmic advice. The results show that different aspects of transparency facilitate distinct trusting beliefs, but only competence belief affects the use of algorithmic advice. It also fully mediates the positive effects of performance and process transparency on the use of algorithmic advice.

This paper thus makes three theoretical contributions. First, we advance the research on algorithmic decision support by shedding light on the role of transparency in promoting the use of algorithmic advice. Drawing on the seminal literature on the performance, process, and purpose attributions of technology [32,38], we further consider three aspects of transparency and unravel their heterogeneous effects in promoting the use of algorithmic advice. Previous studies have mainly considered a single aspect of transparency such as performance and process. By additionally incorporating purpose transparency, we provide holistic insights into the impact of algorithmic transparency. Second, drawing on the knowledge-based trust perspective [35–37], we unravel how different aspects of transparency promote the use of algorithmic advice. By linking three aspects of transparency to trusting beliefs, we examine the mediating roles of different trusting beliefs. Third, we provide a nuanced and valuable understanding of trust in algorithms, which is gaining increasing attention in the literature. Despite the abundance of evidence supporting that people ascribe social rules and expectations to technology [39], previous studies on people's responses to algorithmic advice have rarely viewed algorithms as social actors. Grounded on the literature on social responses to technology [37,40,41], we apply trust, an interpersonal concept, to understand the relationship between transparency and the use of algorithmic advice and unravel the heterogeneous roles of different trusting beliefs in shaping

users' attitudes towards algorithmic advice, enriching the literature on trust in algorithms and our understanding of how users interact with algorithms.

This study also provides valuable implications for practice. We demonstrate that increasing performance and process transparency can help users integrate algorithmic advice into their decision-making. Therefore, disclosing the performance and process information of an algorithm can be an important way for its developers and managers to gain competitive advantages. As algorithms play a central role in market competition, algorithmic transparency strategies can become important differentiators to attract users and promote user engagement. Moreover, we also provide useful guidance on how to leverage different aspects of algorithmic transparency to enhance people's trust.

2. Literature review

2.1. Response to algorithmic advice

Algorithms are computer agents that apply rule- or non-rule-based methods (e.g., machine learning) to produce an output [42]. Research suggests that they have tremendous potential to improve decision-making [43]. For example, simple linear regression models predict better than human intuitions in forecasting numerical outcomes [44]. Algorithms also outperform human experts in professional tasks, such as diagnosing medical [45] and psychological illnesses [43], evaluating employee performance [17], and predicting various human behaviors [46]. Even in very subjective tasks such as predicting the funniness of jokes, algorithms perform better than humans [11]. Although algorithms have the potential to help people make informed decisions [47], whether they can improve decision quality critically depends on people's willingness to accept algorithmic advice [1,9,22].

Despite the superiority of algorithms, studies suggest that humans can exhibit resistance to algorithmic advice [9,14,15]. For example, people prefer a human forecaster to a statistical algorithm, even if the latter has superior performance, which occurs regardless of whether the human forecaster is others or themselves [14]. In predicting stock prices, advice from a statistical model is also more severely discounted than the same but comes from a human expert [21]. In product assortment, workers exhibit aversion to algorithmic advice [29]. Similarly, ride-sharing drivers also resist recommendations from an algorithm, particularly when such recommendations do not match their experience and contradict peer drivers' actions [48]. Patients in the medical field also prefer physicians who do not consult algorithms before making decisions [10]. They are reluctant to take advice from artificial intelligence, even if its performance is superior [9]. Such a phenomenon is also evident among professionals [49]. Employers are more willing to rely on their judgments rather than recommendations from algorithms [16]. Similarly, auditors are reluctant to accept mechanical decision aids [18]. Although some studies suggest that people sometimes prefer algorithmic advice [30,34], it is only a potential tendency rather than a general attitude. While people are willing to seek support from algorithms, they are often more cautious about actually using algorithmic advice [1,14,22]. Particularly in making subjective or moral decisions, people often exhibit serious resistance to algorithmic advice [1,6,11].

Research has identified several reasons for the resistance to algorithmic advice. People generally hold unrealistic expectations of algorithms and are overconfident about their own competence [14,27,28], believing that algorithms may neglect individual uniqueness and fail to take one's characteristics and circumstances into account [9]. More importantly, several studies suggest that the "black box" nature of algorithms may be an important reason for such resistance [27,28,30]. People often receive advice from algorithms that barely reveal relevant information [30,50,51]. The lack of transparency and understanding about algorithms leads them to resist algorithmic advice in the same way that an ordinary person is reluctant to accept advice from strangers [27,52]. Consequently, scholars have recently begun to investigate the

effect of algorithmic transparency on people's use of algorithmic advice.

2.2. Algorithmic transparency

Algorithmic transparency refers to the presence of information related to an algorithm, which helps decision-makers to better evaluate its advice [26,30]. Previous research has proposed a 3P (performance, process, and purpose) framework to classify the key information bases for users' evaluation of and reliance on a technology [32,38,53]. Specifically, performance information illustrates what the technology does, process information describes how the technology operates, and purpose information relates to why the technology is designed [32,53]. This 3P framework provides a more comprehensive and detailed understanding of algorithmic transparency, namely that increasing algorithmic transparency involves disclosing information about the performance, process, and purpose of an algorithm. Following this taxonomy, we focus on three aspects of algorithmic transparency: performance, process, and purpose transparency. Performance transparency typically refers to the disclosure of information about the accuracy or past performance of an algorithm, process transparency emphasizes information about the underlying rationale or working process of an algorithm, and purpose transparency describes information about the motivation and goal of an algorithm [32]. Based on our review, the 3Ps are also the most prominent aspects of interest in the current literature on algorithmic transparency (see Table 1).

Increasing *performance transparency* is related to informing people about the performance and competence of an algorithm in performing certain tasks [30,32,57]. Several studies have investigated the effect of performance transparency in medical settings [9], virtual teams [56], prediction tasks [59], low-stake (such as choosing movies and books), and high-stake (such as saving money) decisions [57]. Barring a few studies [9,30], most studies suggest that performance transparency enhances people's use of algorithmic advice [6,57]. For example, providing the stated and actual accuracy of an algorithm encourages people to use its advice [54]. Informing the performance also improves people's attitudes towards algorithmic advice in both objective and subjective tasks [1]. Even in moral contexts, where people show strong resistance to algorithms, presenting information about the superior accuracy of an algorithm enhances its acceptance [6].

Increasing *process transparency* enhances people's understanding of how an algorithm arrives at a particular decision, which is deemed an essential aspect of algorithmic transparency [11,26,32]. Similar to performance, process transparency has also been shown to positively affect people's attitudes towards algorithmic advice. For example,

explaining how a recommendation agent works alleviates the information asymmetry and the associated uncertainty between the agent and users [37]. Yeomans et al. [11] conducted controlled experiments in which participants were asked to evaluate the funniness of jokes using algorithms. They found that participants showed a greater preference for algorithmic advice that was presented with process information. Similarly, Lehmann et al. [26] found that providing process explanations for a complex algorithm increased the use of its advice. They also reported two practical cases suggesting that people prefer algorithms that reveal underlying process information.

Besides performance and process, *purpose transparency* (i.e., the presence of purpose information about an algorithm) may also influence people's attitudes [30,32,37], but it has rarely been empirically examined. Algorithms, being impersonal, do not have their own intentions, but they carry the purpose of their designers or providers (e.g., why an algorithm is designed and for whose benefit) [32,37]. Particularly in recent times, when algorithms have been increasingly used for manipulation [60], their trustworthiness has also been called into question. Algorithms may be designed to act in a way that increases their owner's business, rather than being completely unbiased in their recommendations [60–62]. Without purpose information, people are more likely to suspect that the algorithm is working to increase the benefit of its providers rather than themselves [37], and the research on advice-taking also suggests that people may not accept advice if they perceive the advisor to be giving advice out of self-interest [52]. Therefore, it's also important to investigate the role of purpose transparency in enhancing the use of algorithmic advice.

In summary, emerging studies highlight the importance of algorithmic transparency [24,27,28]. However, there are two notable gaps in the current literature. First, previous studies have mainly focused on how a single aspect of algorithmic transparency (either performance or process) will affect the use of algorithmic advice, but they have rarely examined the role of purpose transparency. Furthermore, there is little empirical research that incorporates different aspects of transparency (i.e., the 3Ps) and holistically investigates their effects on people's attitudes towards algorithmic advice. This issue matters since it's important to consider the heterogeneity and relative effectiveness of different aspects of transparency [32]. Second, most studies have focused on the direct effect of transparency on the use of algorithmic advice, but the underlying mechanism has rarely been examined. Thus, we aim to address these issues by simultaneously investigating the effects of three aspects of transparency (performance, process, and purpose) and exploring the underlying mechanism.

2.3. The theory of knowledge-based trust

According to the “computers as social actors” view [39], people often unconsciously apply social rules and expectations to technological artifacts and treat them like real human beings rather than simple tools [37,41]. The social nature of technologies enables people to form social perceptions towards them and thus treat them as recipients of trust, a concept that originally developed from interpersonal relationships [41]. Moreover, the components of human-technology trust are not very different from those of interpersonal trust, and people also use words such as “honesty”, “integrity”, and “friendship” to describe a technological artifact [63]. Recent information systems research has incorporated trust into the study of human-technology interaction and demonstrated its fundamental role in shaping users' attitudes [37,40,41]. For example, trust often determines users' adoption of decision-support technologies, such as recommendation systems [37,41]. In addition, trust is also important in the process of taking advice, especially when it is difficult to objectively assess the usefulness of advice, the acceptance or rejection of advice will largely depend on how much the advisor is trusted [52].

Given the applicability and importance of trust in the context of algorithmic advice, we adopt the theoretical perspective of knowledge-

Table 1
Representative research on algorithmic transparency.

Source	Aspects of algorithmic transparency		
	Performance	Process	Purpose
Turel & Kalhan [22]	✓		
Bigman & Gray [6]	✓		
Longoni et al. [9]	✓		
Castelo et al. [1]	✓		
Yin et al. [54]	✓		
Clegg et al. [55]		✓	
Dennis et al. [56]	✓		
You et al. [30]	✓		✓ (Mentioned but not tested)
Liu et al. [48]	✓		
Dietvorst et al. [14]	✓		
Saragih & Morrison [57]	✓		
Yeomans et al. [11]		✓	
Lehmann et al. [58]		✓	
Poursabzi et al. [31]		✓	
Lehmann et al. [26]		✓	
The Present Study	✓	✓	✓

based trust to understand the mechanism through which transparency affects the use of algorithmic advice [36,37]. Knowledge-based trust posits that trust is based on the knowledge and predictability of a trustee's behavior and characteristics [36]. When people know more about a trustee, they can more easily and accurately predict his or her thoughts and reactions, which ultimately contributes to trust [36]. Since transparency implies the disclosure of information about an algorithm, it stimulates users' knowledge-based reasoning, which increases their understanding of the algorithm [64]. Transparency also increases the predictability of the algorithm's utility and behavior, thus contributing to the building of knowledge-based trust [37,60,65]. Conversely, opaque algorithms may hinder the building of knowledge-based trust and thus reduce the use of algorithmic advice [24,25]. Previous research on recommendation systems has also considered transparency (e.g., the provision of explanations) as a knowledge-based trust-building mechanism, because explanations transfer knowledge to users about what, how, and why the system does, thus bridging the "knowledge gap" between users and the system, and removing the information asymmetry obstacle to building trust [37,65]. Therefore, the knowledge-based trust perspective provides a theoretical basis for understanding the role of algorithmic transparency.

Considering that the three aspects of transparency (i.e., performance, process, and purpose) provide different information about an algorithm, they also differ in the way they facilitate users' knowledge-based trust. We further adopt the trust model proposed by Mayer et al. [66] to disentangle the underlying knowledge-based trust-building mechanisms through which the three aspects of transparency affect the use of algorithmic advice. In this model, trust is conceptualized as a positive perception that includes three trusting beliefs: competence, integrity, and benevolence [66,67]. The three trusting beliefs apply to this study as users' trust in technology is also based on evaluations of its competence, integrity, and benevolence [68]. Specifically, competence refers to the trusting belief that an algorithm has the ability to perform a particular task, integrity refers to the trusting belief that an algorithm adheres to a set of acceptable principles, and benevolence refers to the trusting belief that an algorithm is good for users without a profit motive [65]. From the knowledge-based trust perspective, three aspects of transparency convey knowledge about an algorithm's performance, process, and purpose, respectively, based on which users can evaluate and predict its capabilities, workings, and intentions, and thus build corresponding trusting beliefs. Previous research has also shown that different types of explanations about a decision support system lead to different trusting beliefs [37]. Therefore, we examine the three trusting beliefs to clarify the mechanisms through which different aspects of transparency affect the use of algorithmic advice.

3. Research model and hypotheses

Drawing on the knowledge-based trust perspective [36,37], we

establish the links between three aspects of transparency and the use of algorithmic advice, which are mediated by different trusting beliefs. Fig. 1 illustrates the research model. Overall, increasing transparency can positively affect the use of algorithmic advice by facilitating users' knowledge-based trust, and the intermediate knowledge-based trust involves multidimensional trusting beliefs (i.e., competence, integrity, and benevolence) that arise from the knowledge and predictability of various features of an algorithm [25,32]. Thus, the three aspects of transparency differ in the trust-building pathways that affect the use of algorithmic advice, depending on the type of knowledge about an algorithm that people can gain from the disclosed information. Next, we will separately discuss the effects of different aspects of transparency on the use of algorithmic advice, as well as the corresponding mediation of trusting beliefs.

3.1. The impact of performance transparency on the use of algorithmic advice

The first aspect of algorithmic transparency we consider is performance transparency. The presence of information about the performance of an algorithm is likely to positively affect the use of algorithmic advice by enhancing users' competence belief, which refers to the perception that an algorithm has the capability, skills, and expertise to perform effectively [25,32]. From the knowledge-based trust perspective [36,37], competence belief is facilitated when people know about how well an algorithm will perform, which can be explicitly inferred from its performance information. When receiving advice from an algorithm without any performance cues, it is difficult for users to assess how well the algorithm is performing as its effectiveness and expertise are less predictable. In contrast, increasing performance transparency of an algorithm (e.g., disclosing its accuracy) allows users to gain insight into its performance and calibrate their confidence in its ability to support decisions. Therefore, considering the abundant evidence demonstrating the superior performance of algorithms [17,19,43], increasing performance transparency may facilitate users' competence belief. Furthermore, since advisors' competence and expertise are often strongly related to the quality of their advice, if users perceive an algorithm to be competent and skilled, they will have higher evaluations of its advice and thus be more likely to accept it [52]. Previous studies have also suggested that presenting the performance information about an algorithm can promote the use of its advice [1,54,57], even in domains where human advisors are preferred [1,6]. Based on the above discussion, performance transparency is supposed to enhance the use of algorithmic advice by facilitating users' competence belief. Therefore, we hypothesize:

H1a. Performance transparency is positively related to the use of algorithmic advice.

H1b. Competence belief mediates the relationship between

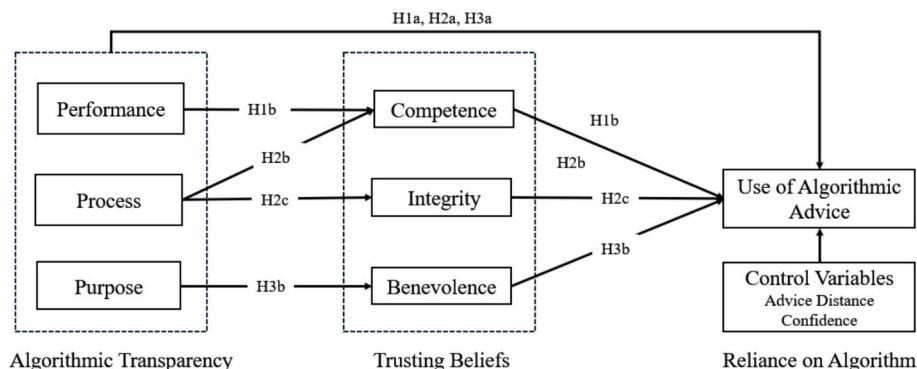


Fig. 1. Research model.

performance transparency and the use of algorithmic advice.

3.2. The impact of process transparency on the use of algorithmic advice

Process transparency is also deemed an essential aspect of algorithmic transparency, which can be increased by revealing the underlying working process of an algorithm [26,32]. We suppose that process transparency can positively affect the use of algorithmic advice by facilitating two trusting beliefs: competence and integrity. Unlike human advisors who can explain the rationale behind their advice, algorithms are often black boxes that lack adequate explanations of their underlying working process, making their advice seem unreliable and thereby leading to users' resistance [27]. Previous research has shown that people can not only assess the competence of a technology directly from its performance information but can also infer its competence from the underlying working process [32,37]. For example, the operating rule of a recommendation system is a strong cue for inferring its competence, and users who receive more process information about the system will perceive it as more competent [65]. Therefore, process transparency increases the predictability of an algorithm's performance to some extent and is supposed to facilitate users' competence belief, which in turn promotes the use of algorithmic advice.

In addition, integrity belief, which refers to the perception that an algorithm adheres to a set of acceptable principles and rules (e.g., being honest and keeping promises), can also be evoked by increasing process transparency [25,32,60]. From the knowledge-based trust perspective [35,36], to enhance integrity belief, there should be available cues to indicate that the algorithm is honest and fair, which can be inferred from its operating process. In practice, algorithms are sometimes manipulated to produce biased or unfair outputs, which may raise concerns regarding their integrity. Disclosing an algorithm's process information can help alleviate such concerns, as it shows how the algorithm generates advice, and thus users are more likely to know its openness and fairness. Therefore, process transparency is supposed to facilitate integrity belief. Furthermore, like taking advice from human advisors, integrity belief also plays a critical role in the use of algorithmic advice [60]. When users perceive an algorithm to be honest and unbiased, they are also more likely to accept its advice. Based on the above discussion, process transparency can positively affect the use of algorithmic advice through the joint mediation of competence and integrity beliefs. Therefore, we hypothesize:

H2a. Process transparency is positively related to the use of algorithmic advice.

H2b. Competence belief mediates the relationship between process transparency and the use of algorithmic advice.

H2c. Integrity belief mediates the relationship between process transparency and the use of algorithmic advice.

3.3. The impact of purpose transparency on the use of algorithmic advice

Purpose transparency, which describes the presence of purpose information of an algorithm [25,32], is also likely to positively affect the use of algorithmic advice. We suppose this positive effect to be mediated by benevolence belief, which refers to the perception that an algorithm has a positive orientation towards its users, such as caring about users' needs and acting in their interests [37]. From the knowledge-based trust perspective [35,36], to improve benevolence belief, there should be available information to confirm that the algorithm has good intentions towards users, which can be judged by its purpose (e.g., why it is designed and who it is intended to benefit). As noted earlier, information asymmetry may induce opportunistic behavior by an algorithm's developers, raising concerns about the goal incongruence between the algorithm and users [37]. Users may suspect that the algorithm only works for certain entities rather than acting in their interests if they

know nothing about its purpose, resulting in a low level of benevolence belief [37]. In this context, purpose information can demonstrate that the algorithm is designed to fulfill users' needs and to convey the goodwill of the providers [37]. Acknowledging the good intentions of the algorithm can address the potential concerns about goal incongruence, and make users perceive that it cares about their interests [37,65]. Therefore, purpose transparency enhances the predictability of an algorithm's good orientation and is supposed to facilitate users' benevolence belief. Furthermore, benevolence belief is also important for advice acceptance [52,69]. Previous research has suggested that the perceptions of advisors' intentions influence the use of their advice. When people perceive advisors as more benevolent (e.g., a non-profit advisor vs. a for-profit advisor), they are more likely to accept the advice [69]. Accordingly, users are more willing to accept the algorithmic advice if they have a higher level of benevolence belief in it. Based on the above discussion, purpose transparency can positively affect the use of algorithmic advice by facilitating benevolence belief. Therefore, we hypothesize:

H3a. Purpose transparency is positively related to the use of algorithmic advice.

H3b. Benevolence belief mediates the relationship between purpose transparency and the use of algorithmic advice.

Table 2 summarizes the relationships between the three aspects of transparency and the corresponding trusting beliefs. We only hypothesize these relationships and exclude the other possible links between transparency and trusting beliefs (i.e., performance transparency and integrity/benevolence beliefs, process transparency and benevolence belief, and purpose transparency and competence/integrity beliefs) for the following reasons. First, according to Mayer et al.'s trust model, although the three trusting beliefs are in the same basket, they capture unique dimensions of trust and can be considered separately [66]. For example, an algorithm may be considered competent but not honest or benevolent [37]. Second, from the knowledge-based trust perspective, each aspect of algorithmic transparency conveys only limited knowledge, based on which users are unable to fully understand and predict the algorithm. For example, an algorithm with superior performance may also be manipulated by its developers to provide biased and malicious advice to profit from users. Thus, although performance transparency can increase the predictability of an algorithm's ability, users still can't judge whether it is following a set of acceptable principles and acting in their interests [65]. In this case, only one trusting belief - competence - can be effectively evoked. Similarly, although users can infer an algorithm's competence and integrity from its process information, they know little about the algorithm's true intention in providing advice, and therefore process transparency can hardly build benevolence belief. In addition, as purpose transparency does not provide any information to predict an algorithm's ability or fairness, it may not explicitly facilitate users' competence and integrity beliefs [25,37]. In summary, considering that different aspects of transparency mainly enhance knowledge and predictability of the different features of an

Table 2
Relationship between transparency and trusting beliefs.

Aspects of transparency	Trusting beliefs	Logic
Performance	Competence	Performance information enhances people's trust in the ability of an algorithm to work effectively in certain situations.
Process	Competence and Integrity	Process information enhances the perception that an algorithm is based on sound logic and can provide unbiased advice.
Purpose	Benevolence	Purpose information enhances the perception that an algorithm is designed to benefit its users.

algorithm, we only hypothesize these salient “algorithmic transparency-trusting belief-use of algorithmic advice” mediating effects as discussed above but will explore other possibilities in the data analysis section.

4. Research method

To test our hypotheses, we designed a 2 (performance transparency: high vs. low) $\times$ 2 (process transparency: high vs. low) $\times$ 2 (purpose transparency: high vs low) between-subjects experiment.

4.1. Experimental design

To examine the influence of increasing transparency on people’s use of algorithmic advice, we designed an experiment based on the “judge-advisor system” (JAS) paradigm [1,26,34], which simulates the algorithmic advice-assisted decision-making situation. Traditionally, the JAS design requires participants to make decisions with the help of an “advisor.” At the beginning, participants are required to complete a decision task (e.g., predicting product sales). They are asked to make decisions themselves first, yielding their initial judgments. Then, they are advised by an advisor. Since the participants can freely determine whether or how much to alter their initial judgments after receiving advice, researchers can thus compare the difference between the initial and final decisions and quantify participants’ utilization of the advice. We used this JAS paradigm but set an algorithm as the advisor instead of a human.

We designed a prediction task requiring participants to forecast the sales volume of three products¹ based on historical information. Participants were given monthly sales information for each product as well as the local GDP trend over the past four years. They were required to forecast each product’s sales in January of the fifth year. Based on the given information, participants then made their initial forecasts. They were then informed that an algorithm had also made predictions and were allowed to revise their initial forecasts. To mimic the algorithmic advice, we adopted the forecast results of a long short-term memory (LSTM) algorithm as the advice.² In other words, the LSTM algorithm was the “advisor.” The algorithm’s name was not revealed to participants to avoid potential bias. All participants received the same advice and only knew it came from an algorithm.

In addition, to manipulate the three aspects of algorithmic transparency (performance, process, and purpose), we provided different types of explanations to participants when they received algorithmic advice, which is a common way to manipulate algorithmic transparency [26]. Specifically, regarding the performance explanation, we provided performance explanations based on the algorithm’s prior performance on similar tasks. For process explanations, we offered a basic explanation of the working process of the LSTM algorithm. Finally, we described the purpose of this algorithm as the purpose explanation. The three explanations are presented in Table 3. We combined these explanations and manipulated their provisions, resulting in 8 groups. Illustrations of our manipulations and prediction tasks are presented in Appendix B.

¹ The three products are sticker, hatter and mut which are common in lives. It allows us to avoid potential bias resulted from participants’ domain knowledge.

² We created fictional sales information for the three products based on a Kaggle project. The LSTM forecasting results were also adopted from a contender’s submission: see Walter Reade. (2022). Tabular Playground Series - Jan 2022. Kaggle. <https://kaggle.com/competitions/tabular-playground-series-jan-2022>

Table 3

Explanations in experiment manipulation.

Aspects of transparency	Explanations description
Performance	Based on previous performance, the algorithm can achieve over 90% accuracy in completing similar tasks and has a 97% possibility of making forecasts that deviate only 5% from the actual value. The algorithm constructs a forecasting model by calculating the pattern of sales change using time series analysis. It then corrects and adjusts the forecasting model, combining the change in GDP. It finally estimates the sales volume of the three products in January of the 5th year.
Process	The algorithm is designed to help decision-makers forecast product sales and thus make better decisions and strategies.
Purpose	

4.2. Participants and procedures

We recruited participants from Credamo to join the experiment. Credamo³ is one of the most popular online platforms that provide data collection services in China. It is pretty similar to Amazon’s Mechanical Turk. Credamo sent our links to invite its registered members to join the experiment. To ensure the quality of collected data, we adopted the premium sampling service offered by Credamo. Only members who performed well in previous responses, such as having high credit scores ranked by Credamo and over 80% adoption rate, were recruited. In total, we recruited 480 participants. We rewarded each participant with 3 RMB to encourage them towards serious participation.

At the beginning of the experiment, participants reported their demographics and rated their trust propensity. They were then randomly assigned to one of the eight groups, where they were required to forecast future sales of three products based on historical information, as described in the previous section. We also incentivized participants to treat the task seriously by informing them that those who made more accurate forecasts would receive extra bonuses. Participants in all eight groups conducted the same prediction tasks and reported their confidence in initial predictions, after which they received identical algorithmic advice. The only difference between groups was whether they also received explanations related to the algorithm (i.e., performance, process, and purpose information). After completing the final forecasting tasks, participants rated their trusting beliefs in the algorithm and three manipulation check items. Fig. 2 presents the experiment procedures.

In the JAS paradigm, the relative shift of participants’ initial and final decisions accurately reflects how much the advisors’ advice influences participants. Specifically, we calculated each participant’s weight on advice (WOA) based on the following formula:

$$WOA = \frac{\text{Initial forecast} - \text{Final forecast}}{\text{Initial forecast} - \text{Algorithmic advice}}$$

Each participant’s average WOA over three products was used to measure their use of algorithmic advice. In most cases, the WOA value would range from 0 to 1. The WOA value of 0 meant that participants ignored the algorithmic advice, while a WOA value of 1 implied that participants fully followed the algorithmic advice. Thus, any value between 0 and 1 reflected the extent to which participants accepted the algorithmic advice. Based on each participant’s average WOA value over three products, we excluded participants whose values were <0 or >1 since these outliers might undermine the relative power of normal samples and induce potential bias. Those participants whose initial forecasts were the same as the algorithmic advice were also removed because if the initial forecasts equaled the algorithmic advice, the WOA

³ Please see <https://www.credamo.com/home.html#/> and https://help.credamo.com/web/#/?page_id=118 for papers that use Credamo’s data collection service.

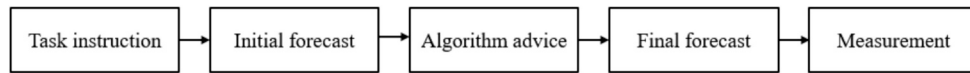


Fig. 2. Experiment procedure.

value could not be calculated, and the participants also did not need to decide whether to use the algorithmic advice. Finally, 443 valid responses were left, meeting the required sample size ($N = 210$) to ensure sufficient statistical power (around 95%) at the significant level of 5% and the medium effect size ($f = 0.25$), as calculated by the G*Power software (v 3.1.9.7). The final sample distribution of each experiment group is shown in Table 4.

Participants' demographic information is presented in Table 5. Nearly half of the participants were female (59.6%), and most of them were between 20 and 30 years old (59.8%) and had a bachelor's degree (72.7%). The mean values and standard deviations of variables in each group are presented in Table 6. We also ensured that there was no significant difference in participants' age, gender, and level of education among different groups.

4.3. Measurement

We adapted previously validated measurement items to our contexts (Appendix A). Specifically, we adapted measures for the three trusting beliefs and trust propensity from previous research on trust in technology [37,70]. Items for confidence in initial predictions were adapted from Tost et al. [71]. The measurement items were rated on a seven-point Likert scale ranging from "strongly disagree" to "strongly agree." We also tested the reliability and validity of these constructs. The Cronbach's alpha values ranged from 0.721 to 0.871, and all items had a significant and higher loading (ranging from 0.713 to 0.896) on corresponding constructs [72]. We also calculated the average variance extracted (AVE) values and ensured each construct's AVE was larger than the squared correlation with other constructs [73]. Thus, we confirmed the reliability and validity of our constructs.

5. Data analysis and results

5.1. Manipulation check

We conducted a manipulation check to test the effectiveness of our manipulations and found that providing explanations increased corresponding aspects of algorithmic transparency. Specifically, we found that participants in groups with performance explanations rated their understanding of algorithmic performance significantly higher than those without explanations (Mean difference = 0.65, $p < 0.001$). This was also true for groups with process explanations (Mean difference = 0.95, $p < 0.001$) and purpose explanations (Mean difference = 0.56, $p < 0.001$). Thus, our manipulations were successful.

Table 4
Sample distribution.

	Without process explanation	With process explanation
Without performance explanation	Without purpose explanation (group 1) $N = 55$	Without purpose explanation (group 5) $N = 54$
	With purpose explanation (group 2) $N = 57$	With purpose explanation (group 6) $N = 57$
With performance explanation	Without purpose explanation (group 3) $N = 57$	Without purpose explanation (group 7) $N = 54$
	With purpose explanation (group 4) $N = 55$	With purpose explanation (group 8) $N = 54$

Table 5
Demographics information.

Items	Frequency	Percentage
Gender	Female	264
	Male	179
Education	Senior high school or below	7
	Junior college	24
	Bachelor	322
	Undergraduate	84
	Doctor	6
	Below 20	6
Age	20–25	148
	26–30	117
	31–36	121
	37–45	33
	Above 45	18

Table 6
Descriptions of variables.

Group	TP	COF	COM	INT	BEN	WOA
1	5.61 (0.56)	5.34 (0.83)	5.20 (1.16)	5.56 (1.01)	4.83 (1.25)	0.45 (0.26)
2	5.56 (0.50)	5.40 (0.83)	5.53 (0.71)	5.76 (0.74)	5.35 (0.91)	0.51 (0.30)
3	5.66 (0.57)	5.55 (0.88)	5.56 (0.78)	5.84 (0.76)	5.43 (1.13)	0.51 (0.24)
4	5.63 (0.58)	5.49 (0.88)	5.83 (0.56)	5.99 (0.55)	5.39 (0.90)	0.56 (0.24)
5	5.47 (0.76)	5.27 (0.99)	5.72 (0.79)	5.85 (0.65)	5.36 (0.88)	0.57 (0.26)
6	5.66 (0.50)	5.35 (0.90)	5.79 (0.60)	5.90 (0.70)	5.03 (1.22)	0.61 (0.21)
7	5.71 (0.55)	5.22 (1.01)	5.72 (0.52)	5.92 (0.60)	5.36 (0.90)	0.51 (0.24)
8	5.65 (0.58)	5.31 (1.03)	5.85 (0.44)	5.88 (0.56)	5.20 (1.25)	0.56 (0.29)

Note: TP = trust propensity; COF = confidence in initial forecasts; COM = competence belief; INT = integrity belief; BEN = benevolence belief; WOA = weight on advice. Standard deviations are in parentheses.

5.2. Effects of algorithmic transparency on WOA

In the following sections, we adopted the WOA as the dependent variable to examine how three manipulated variables related to algorithmic transparency affect the use of algorithmic advice. We first conducted an ANCOVA with confidence in initial predictions and advice distance⁴ as covariates. The results are shown in Table 7 and Fig. 3. We found that performance ($F = 3.947$, $P < 0.05$) and process transparency ($F = 4.804$, $P < 0.05$) significantly increased WOA, but purpose transparency did not ($F = 0.200$, $P > 0.1$). Thus, H1a and H2a were supported, but H3a was not.

Interestingly, we also found a significant interaction effect between process and purpose transparency ($F = 6.378$, $P < 0.05$). As shown in

⁴ Previous research suggests that the distance of advice (the difference between advice and initial judgment) may influence advice taking, especially when the advice is quite different from their own judgment [52]. Since participants in our study received three pieces of advice, we calculated the difference rate of each advice, that is the absolute value of difference between advice and initial forecasts divided by the value of advice, and used the average difference rate as the advice distance.

Table 7
Results of ANCOVA: effects of transparency on WOA.

Source	Type III sum of squares	Degrees of freedom	Mean square	F	P-value
Covariates					
Confidence	0.352	1	0.349	5.525	0.019
Advice Distance	0.726	1	0.726	11.397	0.001
Main effect					
Performance	0.251	1	0.251	3.947	0.048
Process	0.306	1	0.306	4.804	0.029
Purpose	0.013	1	0.013	0.200	0.655
Interaction effect					
Performance×Process	0.008	1	0.008	0.127	0.721
Performance×Purpose	0.004	1	0.004	0.063	0.803
Process×Purpose	0.406	1	0.406	6.378	0.012
Performance×Process×Purpose	0.005	1	0.005	0.072	0.788

Fig. 4, when process transparency was low, purpose transparency could promote WOA ($P < 0.05$). Nevertheless, this positive effect became insignificant in cases of high process transparency ($P > 0.1$). In other words, increasing process transparency weakened the promoting effect of purpose transparency on WOA. Increasing both purpose and process transparency did not enhance the use of algorithmic advice. For process transparency, despite the significant main effect ($F = 4.804$, $P < 0.05$), its promoting effect also became insignificant when purpose transparency was high ($P > 0.1$) (Fig. 5).

5.3. Mediation analysis

To test the mediating effects of trusting beliefs, we adopted the PROCESS macro [74]. In the mediation model, we treated the three aspects of transparency as the independent variables, corresponding trusting beliefs as the mediators, and WOA as the dependent variable. We also included confidence in initial predictions and advice distance as

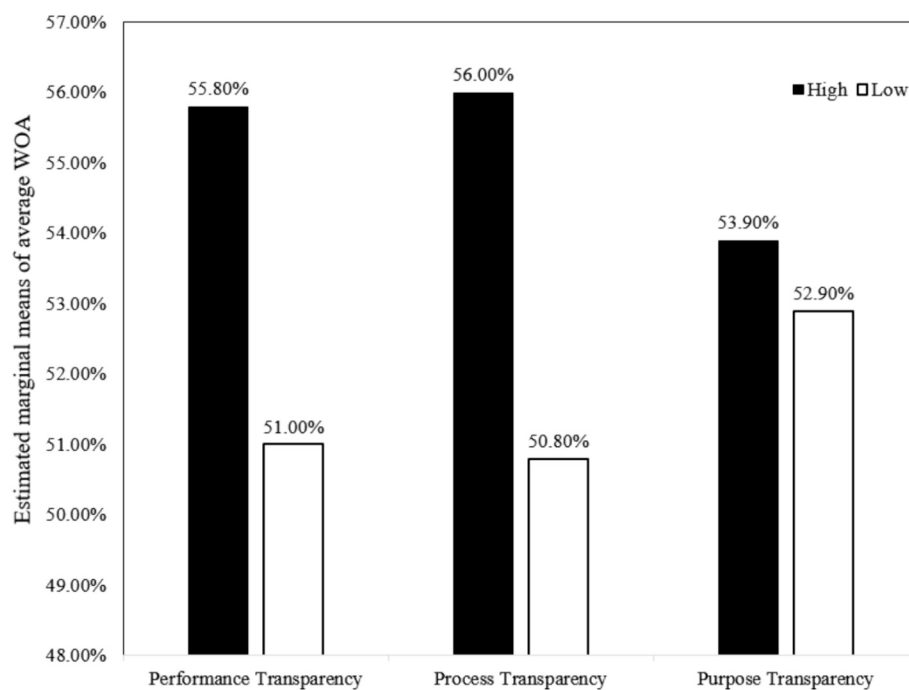


Fig. 3. Estimated marginal means of average WOA.

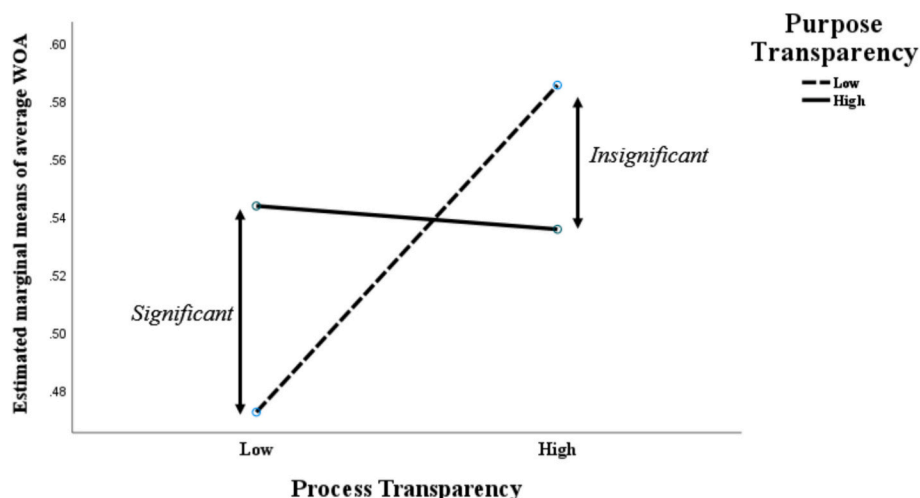


Fig. 4. The effect of purpose transparency.

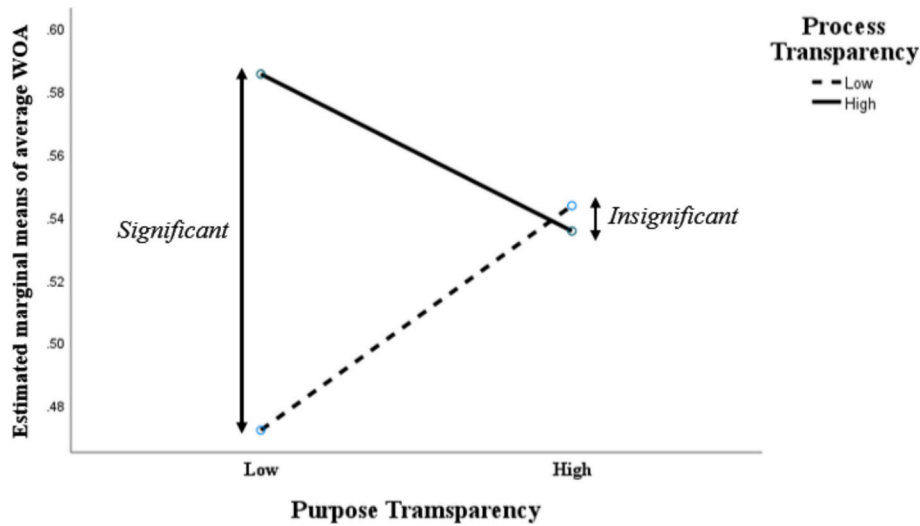


Fig. 5. The effect of process transparency.

covariates. We then performed mediation analyses via bootstrapping ($N = 5000$). Regarding performance transparency, we found that performance transparency promoted competence belief ($\beta = 0.213, P < 0.05$), and the latter significantly affected WOA ($\beta = 0.198, P < 0.001$). Considering the insignificant direct effect of performance transparency ($\beta = 0.141, P > 0.1$), competence belief fully mediated this relationship (95% CI = [0.0057, 0.0950]). Thus, H1b was supported. As for process transparency, competence belief was facilitated by process transparency ($\beta = 0.414, P < 0.01$) and significantly and positively affected WOA ($\beta = 0.185, P < 0.01$). Combined with the insignificant direct effect ($\beta = 0.121, P > 0.1$), competence belief also fully mediated the relationship between process transparency and WOA (95% CI = [0.0231, 0.1420]). Although process transparency indeed enhanced integrity belief ($\beta = 0.245, P < 0.01$), the latter could not significantly affect WOA ($\beta = 0.012, P > 0.1$) and the mediation effect was also insignificant (95% CI = [-0.0261, 0.0330]). Therefore, H2b was supported while H2c was not. Finally, we found that purpose transparency facilitated benevolence belief ($\beta = 0.197, P < 0.05$) but it failed to significantly influence WOA ($\beta = 0.065, P > 0.1$), resulting in an insignificant mediation effect (95% CI = [-0.0076, 0.0433]). H3b was thus unsupported. We summarize the results of mediation tests in Table 8.

To further investigate the effects of process and purpose transparency on trusting beliefs, we conducted additional ANCOVA using trust propensity as the covariate. Consistent with previous findings, process transparency positively and significantly affected integrity belief ($F = 3.939, P < 0.05$), and purpose transparency also positively affected benevolence belief ($F = 3.597, P = 0.059$). These findings further confirmed that although process and purpose transparency facilitated integrity and benevolence beliefs, the mediation effect was unsupported due to the insignificant impact of the two trusting beliefs on WOA.

Table 8
Mediation analyses.

Path	Direct effect	Indirect effect	95% Confidence interval		Support?
PER-COM-WOA	0.1407	0.0420	0.0057	0.0950	Yes
PRO-COM-WOA	0.1212	0.0768	0.0231	0.1420	Yes
PRO-INT-WOA	0.1212	0.0028	-0.0261	0.0330	No
PUR-BEN-WOA	0.0296	0.0127	-0.0076	0.0433	No

Note: PER = Performance Transparency; PRO = Process Transparency; PUR = Purpose Transparency; COM = Competence belief; BEN = Benevolence belief; INT = Integrity belief. All direct effects are insignificant. Standardized coefficients are reported.

Moreover, following previous steps, we also conducted additional mediation analyses to investigate other possible mediation effects of the three trusting beliefs by simultaneously treating them as mediators. The results were consistent with previous analyses. Additionally, we found a significant effect of purpose transparency on competence belief ($\beta = 0.197, P < 0.05$), which could significantly affect WOA ($\beta = 0.202, P < 0.001$). Combined with the insignificant direct effect ($\beta = 0.004, P > 0.1$), competence belief fully mediated the effect of purpose transparency on WOA (95% CI [0.0027, 0.0967]). Nevertheless, this effect was relatively weak ($\beta = 0.04$), and the total effect of purpose transparency was also insignificant ($\beta = 0.042, P > 0.1$).

6. Discussion and implications

6.1. Discussion of findings

In this study, we simultaneously investigate the effects of three aspects of algorithmic transparency (performance, process, and purpose transparency) on the use of algorithmic advice. Drawing on the knowledge-based trust perspective, we unravel the mediating roles of trusting beliefs in the relationship between transparency and the use of algorithmic advice. Through an experiment based on 443 participants, we provide empirical support for some hypotheses (see Table 9). Specifically, we find the heterogeneous effects of different aspects of transparency. Performance and process transparency are effective in promoting the use of algorithmic advice, while purpose transparency only works in cases of low process transparency. Regarding the mechanisms, we find that the three aspects of transparency facilitate different trusting beliefs: performance transparency facilitates competence belief, process transparency facilitates both competence and integrity beliefs, and purpose facilitates benevolence belief. However, only competence belief enhances the use of algorithmic advice. It also fully mediates the promoting effects of performance and process transparency on the use of algorithmic advice.

In addition to the hypotheses testing results, there are other interesting findings. First, the promoting effects of process and purpose transparency are complex and mutually dependent. For process transparency, while it can generally increase the use of algorithmic advice, this effect gets weakened in cases of high purpose transparency. Similar cases also occur for purpose transparency, which only promotes the use of algorithmic advice when process transparency is low. Thus, simultaneously increasing process and purpose transparency does not lead to higher acceptance of algorithmic advice. This can be explained from the information overload perspective. Previous research suggests that

Table 9
Summary of results.

Hypothesis	Results	Detailed findings
H1a: Performance transparency is positively related to the use of algorithmic advice.	Supported	Increasing performance transparency can promote the use of algorithmic advice.
H1b: Competence belief mediates the effect of performance transparency on the use of algorithmic advice.	Supported	Competence belief fully mediates the effect of performance transparency on the use of algorithmic advice.
H2a: Process transparency is positively related to the use of algorithmic advice.	Supported	Increasing process transparency can encourage people to use algorithmic advice, but this effect will be weakened in cases of high-purpose transparency.
H2b: Competence belief mediates the effect of process transparency on the use of algorithmic advice.	Supported	Competence belief fully mediates the effect of process transparency on the use of algorithmic advice.
H2c: Integrity belief mediates the effect of process transparency on the use of algorithmic advice.	Rejected	Process explanation can facilitate integrity belief, which fails to impact the use of advice.
H3a: Purpose transparency is positively related to the use of algorithmic advice.	Rejected	Increasing purpose transparency generally does not bring higher reliance on the advice, but it may work in cases of low process transparency.
H3b: Benevolence belief mediates the effect of purpose transparency on the use of algorithmic advice.	Rejected	Purpose explanation can facilitate competence and benevolence beliefs, but the latter can not significantly improve the use of advice.

increasing transparency may induce information overload and lead to users' cognitive overload, undermining the use of algorithmic advice [30,31]. When presented with both process and purpose information, users may find it difficult to evaluate the usefulness of algorithmic advice, inhibiting its adoption. Previous studies have argued that disclosing too much homogeneous information can cause the problem of information overload [30,31]. This study extends it by demonstrating that simultaneously enhancing different aspects of transparency may also lead to information overload, which may suppress their respective promoting effects on the use of algorithmic advice.

Second, regarding the two unsupported mediation hypotheses (i.e., H2c and H3b), we find that process transparency facilitates integrity belief, but the latter fails to affect the use of algorithmic advice. Similar cases also occur for purpose transparency, which enhances benevolence belief, but the latter doesn't enhance the use of algorithmic advice.⁵ These findings provide evidence that when deciding whether to take advice generated by an algorithm, people pay more attention to its competence rather than whether it adheres to a set of acceptable rules or has a positive orientation towards its users. This is reasonable since only when people perceive an algorithm as having the capability and expertise to perform effectively, they can accept its advice to improve decision-making. It also explains why increasing performance and process transparency can significantly influence the use of algorithmic advice (since they convey strong cues to facilitate competence belief). These findings provide insights into the mechanism underlying the effects of transparency: it calibrates users' competence belief and thus promotes the use of algorithmic advice.

⁵ In addition, we also find that competence belief can be evoked by increasing purpose transparency, which suggests that knowing the purpose of an algorithm can also enhance the evaluation of its competency. Nevertheless, this effect is relatively weak ($\beta = 0.04$).

6.2. Theoretical implications

Our research makes the following contributions to research. First, we provide a fine-grained understanding of algorithmic transparency and its impact on the use of algorithmic advice. Previous studies have mainly examined algorithmic transparency as a whole construct or focused on the singular effect of either performance [1,6,30] or process transparency [11,26,31]. Based on a 3P framework of technology reliance [32], we identify three aspects of algorithmic transparency: performance, process, and purpose transparency. The combination of the 3P dimensions with the concept of transparency refines the connotations of algorithmic transparency, and, in particular, fills the gap in research on purpose transparency. Based on this 3P classification, we further discover the heterogeneous effects and interactions of different aspects of transparency on the use of algorithmic advice, shedding light on their relative effectiveness and interdependence. Compared to performance and process transparency, purpose transparency is not as effective for the use of algorithmic advice. Moreover, purpose and process transparency interfere with each other in facilitating the use of algorithmic advice. Thus, while our study provides a comprehensive 3P framework for understanding algorithmic transparency and its impact, it also highlights the specificity of purpose transparency, which can be further investigated by future research.

Second, we reveal the mechanisms underlying the promoting effect of transparency on the use of algorithmic advice. Drawing on the knowledge-based trust theory [35–37], we present a unified theoretical perspective for understanding the effect of algorithmic transparency. Traditional research on interpersonal trust has suggested that it often takes a long period of interaction and mutual understanding between people to build knowledge-based trust [35,36]. However, this study extends knowledge-based trust to the human-algorithm relationships and finds that users' knowledge-based trust in algorithms can be directly induced by algorithmic transparency, even in their initial interaction. By introducing three different trusting beliefs (i.e., competence, integrity, and benevolence) that are associated with three different aspects of algorithmic transparency, we also refine and disentangle the knowledge-based trust-building mechanism in human-algorithm interactions. That is, different aspects of transparency convey different types of knowledge about an algorithm and thus facilitate corresponding trusting beliefs that ultimately affect the use of algorithmic advice. Therefore, we provide in-depth and fine-grained insights to understand the underlying mechanisms of the impact of transparency on the use of algorithmic advice.

Third, we enrich and deepen the existing research on trust in algorithms, which is gaining widespread attention. Trust is a fundamental predictor of users' attitudes and reactions towards a technology [41,75], and previous studies have mainly focused on users' general trust rather than specific trusting beliefs [1,76,77]. We distinguish users' three different trusting beliefs (i.e., competence, integrity, and benevolence beliefs) in an algorithm and find their heterogeneous roles in the use of algorithmic advice. Specifically, competence belief is important for the use of algorithmic advice, but integrity and benevolence beliefs are not. This finding accurately identifies the specific type of trust that determines users' attitudes and behavior towards an algorithm. More importantly, it suggests that simply focusing on the general trust in technology is biased and that there is a need to study trust from a finer-grained perspective, particularly to examine the differences among trusting beliefs. By revealing how the three aspects of algorithmic transparency affect the three trusting beliefs, we also build a bridge between two influential theories of trust, namely Lee and See's technology reliance framework (Performance/Process/Purpose) [32] and Mayer et al.'s integrated trust model (Competence/Integrity/Benevolence) [66], both of which are widely used to study trust in technology. Although previous research has qualitatively discussed the one-to-one correspondence between the two sets of trust-related elements (i.e., performance→competence, process→integrity, and purpose→benevolence) [78], our empirical findings, while supporting the

above inferences, suggest that the trusting belief of a technology's competence is also related to its process and purpose elements (although purpose's effect is relatively weak).

6.3. Practical implications

In light of our findings, we also provide valuable implications for practice. First, while increasing algorithmic transparency is considered an effective strategy for overcoming users' resistance to algorithmic advice, its effect may only hold for certain cases. Our study suggests that only performance and process transparency can generally promote the use of advice, but not purpose transparency. In addition, simultaneously increasing process and purpose transparency has no effect, and may even lead to lower acceptance of algorithmic advice. As demonstrated in our study, the effect of process transparency is weakened in cases of high purpose transparency, and purpose transparency is also only effective when process transparency is low. Therefore, it is important for practitioners to consider the specific aspects of transparency and the potential interactions between them. Specifically, disclosing only the performance information or the working process of an algorithm is the most effective and reliable way to promote users' acceptance of algorithmic advice, and the intention of an algorithm should be disclosed with caution because there can be boundary conditions for its promoting effect.

Second, algorithmic transparency is also effective in facilitating users' trust, which is important in the interaction between humans and algorithms [76,77,79]. However, not all trusting beliefs can ultimately improve users' acceptance of algorithms. Our study demonstrates that of the three trusting beliefs, only competence belief can positively affect the use of algorithmic advice, indicating that users may pay more attention to the expertise and capability of an algorithm when relying on it to make decisions. Only when the algorithm is competent enough to perform a specific task will users use its advice. Therefore, it is crucial to improve users' perceptions of the competence of an algorithm, and disclosing its performance and process information are two effective and feasible alternatives. Despite their ineffectiveness in promoting the use of algorithmic advice, integrity and benevolence beliefs may also be important in a wider range of human-algorithm interaction scenarios. For example, when algorithms are used to make decisions that will affect people's welfare (e.g., evaluating employees or making moral judgments), the morality-related beliefs (i.e., integrity and benevolence), which are inherently low due to the non-human nature of algorithms, may largely determine people's acceptance [6,80]. Moreover, with the growing concerns about algorithmic bias and discrimination, improving people's integrity and benevolence beliefs in algorithms may become critical in the future. In this regard, our study provides feasible alternatives to cope with such issues: increasing process and purpose transparency. In summary, we suggest that practitioners can leverage the promoting effects of different aspects of transparency in shaping different trusting beliefs, thereby eliciting favorable users' attitudes towards an algorithm.

7. Limitations and future research

There are several limitations in our research. First, the generalizability of our results is a critical issue. Although we examine the promoting effect of transparency on the use of algorithmic advice in a fictional prediction task, its application to other contexts, such as moral decision-making or medical diagnosis, requires further examination. More importantly, since many participants have a bachelor's degree, whether our results also hold for participants with low education levels should also be paid attention to by future research. Second, other aspects of transparency can also be further investigated. For example, in addition to the 3P framework in our study, another study has also adopted a

three-stage framework (i.e., input, process, and output) to differentiate the dimensions of algorithmic transparency [64]. Furthermore, as more aspects of transparency are investigated, researchers can also analyze their relative importance and potential combinations to help practitioners find the most effective strategies to improve users' attitudes towards algorithmic advice. Third, while we examine the effect of transparency, the way to increase transparency also matters. In addition to providing explanations, their design and presentation may be equally important [30]. Therefore, it is also worthwhile to investigate whether changes in the design of explanations, for example, simple or complex explanations of performance, process, and purpose information, will differ in users' responses.

8. Conclusions

In this paper, we investigate the impact of increasing three aspects of algorithmic transparency (i.e., performance, process, and purpose transparency) on people's use of algorithmic advice. Based on the "judge-advisor system" paradigm, we design an experiment where participants complete a prediction task with the assistance of algorithmic advice. We find that increasing algorithmic transparency benefits the use of algorithmic advice. However, this does not hold true for purpose transparency, which only enhances the use of algorithmic advice in cases of low process transparency. In addition, the effect of process transparency can also be weakened when purpose transparency is high. Drawing on the knowledge-based perspective, we also confirm the mediated roles of trusting beliefs in the relationship between transparency and algorithmic advice utilization. The results indicate that although the three aspects of transparency can facilitate different trusting beliefs, only competence belief enhances the use of algorithmic advice. It also fully mediates the promoting effect of performance and process transparency. This study thus contributes to the nascent literature on algorithmic decision support by providing a fine-grained insight into the effect of transparency on algorithmic advice utilization and revealing the underlying mechanism from the perspective of trust. In addition, we provide valuable implications for practice, particularly in guiding how to encourage the use of algorithmic advice, which is important to both individual users and algorithm developers.

CRedit authorship contribution statement

Xianzhang Ning: Writing – original draft, Formal analysis, Conceptualization. **Yaobin Lu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Weimo Li:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Sumeet Gupta:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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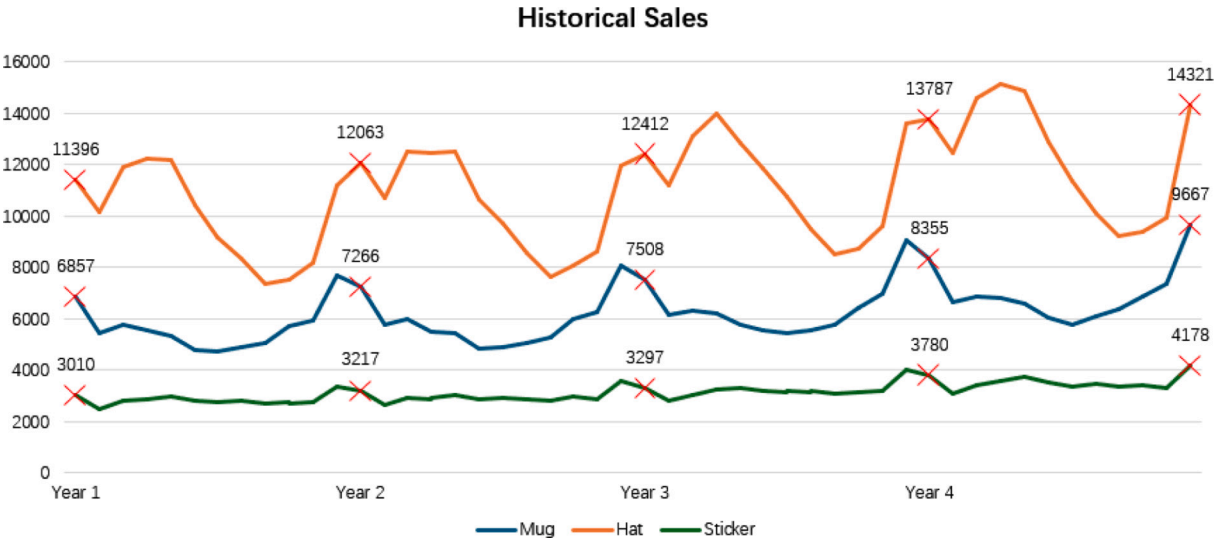
Appendix A. Measurement of variables

Appendix A	
Instrument items.	
Measurement items	Source
Competence belief	(Wang & Benbasat, 2007; Lee & Turbuan, 2001)
This algorithm has the expertise to predict a product's future sales.	
This algorithm has the ability to predict a product's future sales.	
This algorithm has good knowledge about predicting a product's future sales.	
This algorithm is professional in predicting a product's future sales	
Benevolence belief	
This algorithm puts my interest first.	
This algorithm keeps my interests in mind.	
This algorithm tries to understand my needs and interests.	
Integrity belief	
This algorithm provides unbiased predictions.	
This algorithm is honest.	
I consider this algorithm to be of integrity	
Trust propensity	
It is easy for me to trust a person/thing.	
My tendency to trust a people/thing is high.	
I tend to trust a person/thing, even though I have little knowledge of it.	
Trusting something/someone is not difficult for me.	
Confidence	(Tost et al., 2012)
I am very certain about the accuracy of my judgments.	
I feel confident that I performed well on this task.	
I have no doubt my estimates are close to the true values.	

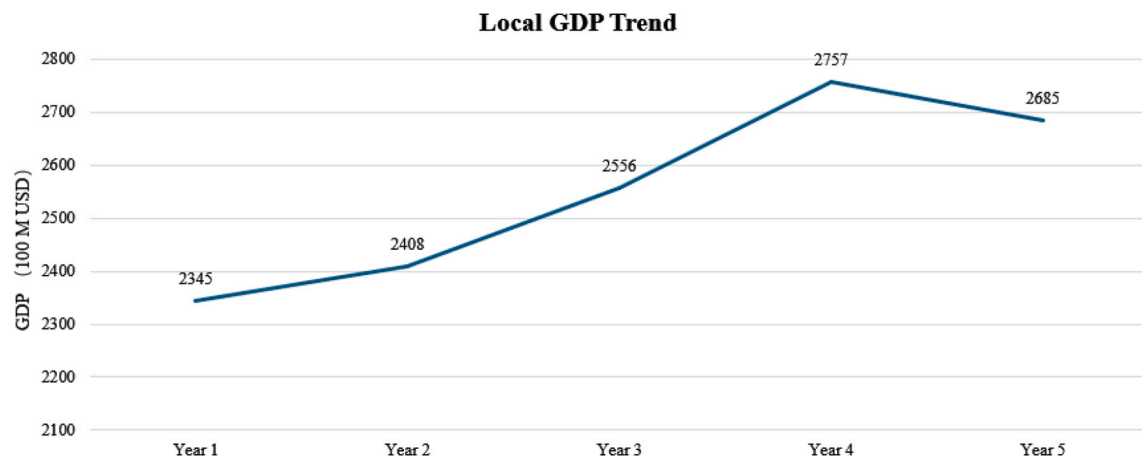
Appendix B. Illustration of experiment task

(Initial forecast)

In this experiment, you are required to predict the sales of three products (i.e., hat, mug, sticker) in a store. Here are the historical sales of the three products over the past four years. Labeled data is the sales of the three products at the beginning of each year, except for the last one (i.e., 14,321,9667,4178), which is the sales information in December of year 4. Please treat this task seriously, we will give extra rewards if your predictions are close to the actual sales!



Since sales are likely to be influenced by local GDP trends, to help you make more accurate predictions, the local GDP trend over the past four years and the GDP prediction in year 5 are also provided.



Now, based on the above information, predict the sales of the three products in January of year 5. Please enter your predictions: Hat: ____; Mug: ____; Sticker: ____.

(Algorithmic advice)

Currently, algorithms have been increasingly used to assist people's decision making, including product sales predictions. An algorithm provided by a tech company has made predictions for the three products. Its predictions are as follows: Hat: 13543; Mug: 8197; Sticker 3663.

Extra explanations: (*example of group 8*).

(1) Based on previous performance, the algorithm can achieve over 90% accuracy in completing similar tasks and has a 97% possibility of making forecasts that deviate only 5% from the actual value.

(2) The algorithm constructs a forecasting model by calculating the pattern of sales change using time series analysis. It then corrects and adjusts the forecasting model, combining the change in GDP. It finally estimates the sales volume of the three products in January of the 5th year.

(3) The algorithm is designed to help decision makers forecast product sales and thus make better decisions and strategies.

(Final forecast)

Now you have a chance to revise your initial forecasts. Please enter your **final predictions**:

Hat: ____; Mug: ____; Sticker: ____.

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